CISS240: Introduction to Programming
Quiz q1402

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. Write a program that gets two integer n and i from the user and prints i–th digit of n:

- if $n = 13524$ and $i = 0$, then 4 is printed,
- if $n = 13524$ and $i = 1$, then 2 is printed,
- if $n = 13524$ and $i = 2$, then 5 is printed,
- if $n = 13524$ and $i = 3$, then 3 is printed,
- if $n = 13524$ and $i = 4$, then 1 is printed,
- if $n = 13524$ and $i = 5$, then 0 is printed,
- if $n = 13524$ and $i = 6$, then 0 is printed,

Here's a test run:

```
13524 2
5
```

Answer:

```
#include <iostream>

int main()
{
    int n, i;
    std::cin >> n >> i;

    int digit = 0;

    if (n < 0)
    {
        n = -n;
    }

    for (int j = 0; j <= i; ++j)
    {
        digit = n % 10;
        n /= 10;
    }
```

```
    std::cout << digit << '\n';
    return 0;
}
```

Q2. Write a program that gets an integer n from the user and continually replace n by the sum of the digits of n until a one digit n is reached. The values taken by n is printed. For instance if the user enters $n = 123456$, then $n = 1+2+3+4+5+6 = 21$, $n = 2 + 1 = 3$. Here's a test run:

```
123456
21
3
```

Answer:

```
#include <iostream>
int main()
{
    int n;
    std::cin >> n;

    int num_digits; // number of digits in n

    // compute num_digits
    num_digits = 0;
    int temp = n;

    while (temp > 0)
    {
        temp /= 10;
        num_digits++;
    }

    while (num_digits > 1)
    {
        int sum_digits = 0;

        // compute sum_digits
        temp = n;
        while (temp > 0)
        {
            sum_digits += temp % 10;
            temp /= 10;
        }

        // print sum_digits
        std::cout << sum_digits << '\n';
```

```
        // get ready for the next iteration
        n = sum_digits;

        // compute num_digits
        num_digits = 0;
        temp = n;
        while (temp > 0)
        {
            temp /= 10;
            num_digits++;
        }
    }

    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.